

Sec 6.4 – The Second Fundamental Theorem of Calculus

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

Not all “nice” functions have “nice” antiderivatives. For example, $f(x) = \cos(x^2)$ does not have an antiderivative made up of constants, powers of x , trigonometric functions, e^x , and $\ln x$.

Problem: How can we find values for an antiderivative of $f(x) = \cos(x^2)$ even when we can't write it as a “nice” function?

If an antiderivative of $f(t) = \cos(t^2)$ exists, $F(t)$, we know from the FTC that

$$F(b) - F(a) = \int_a^b \cos(t^2) dt.$$

Replacing b by x and a by 0 we get

$$F(x) - F(0) = \int_0^x \cos(t^2) dt.$$

If we choose $F(0) = 0$, we get

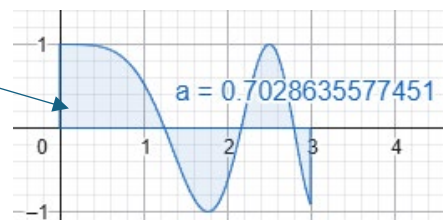
$$F(x) = \int_0^x \cos(t^2) dt.$$

While this doesn't look familiar, this is a function! For every value of x , we can find a value.

For example, using CoCalc, $F(3) = \int_0^3 \cos(x^2) dx \approx .7029$.

Geogebra.org

Integral($\cos(x^3)$, 0, 3)



What is $F(0)$?

Since the limits of integration are the same, $F(0) = \int_0^0 \cos(t^2) dt = 0$

Theorem (Construction Theorem for Antiderivatives)

(Second Fundamental Theorem of Calculus (FTC II))

If f is a continuous function on an interval, and if a is any number in that interval, then the function F defined on the interval as follows is an antiderivative of f :

$$F(x) = \int_a^x f(t) dt.$$

This Construction Theorem, also known as the Second Fundamental Theorem of Calculus, enables us to write antiderivatives of functions that do not have “elementary” antiderivatives.

So, what would an antiderivative of $\frac{\sin x}{x}$ be?

The FTC II gives us, $F(x) = \int_0^x \frac{\sin t}{t} dt$. F is a function, although it may not look like the functions we have seen before.

We can calculate the integral $\int_0^x \frac{\sin t}{t} dt$ to any desired accuracy. This function already has a name, it is called the **sine-integral** and symbolically is denoted $Si(x)$.

So, $Si(x) = \int_0^x \frac{\sin t}{t} dt$, what would the derivative of $x \cdot Si(x)$ be?

We're taking the derivative of the product of two functions, so we should use the Product Rule:

$$\begin{aligned} \frac{d}{dx}(x \cdot Si(x)) &= \left(\frac{d}{dx}x\right)Si(x) + x\left(\frac{d}{dx}Si(x)\right) \\ &= 1 \cdot Si(x) + x\left(\frac{\sin x}{x}\right) \\ &= Si(x) + \sin x \end{aligned}$$

Partner up

- ① Find the integral $\int_1^x t^3 dt$ using the FTC I. Then take the derivative.

(The result is the answer to $\frac{d}{dx} \int_1^x t^3 dt$.)

- ② Use the pattern from #1 to find $\frac{d}{dx} \int_3^x e^t \sin t dt$.

- ③ Find $\frac{d}{d\theta} \int_{-2}^{\theta} \arctan(x^2 + x) dx$

- ④ Find $\int_1^{x^2} \cos t dt$ using the FTC I. Then take the derivative. What connections do you see between the original integral and your final answer?

- ⑤ Use the pattern from #4 to find $\frac{d}{dx} \int_3^{\sin x} e^{-t^2} dt$

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1. Find $\frac{d}{dx} \int_1^x t^3 dt$. First find the integral using the FTC I. Then take the derivative.

By FTC I, $\int_1^x t^3 dt = \frac{t^4}{4} \Big|_1^x = \frac{x^4}{4} - \frac{1^4}{4}$. Then we take the derivative to get $\frac{d}{dx} \int_1^x t^3 dt = x^3$. That is, we get back the integrand with the variable from the upper limit. Notice that changing the lower limit to another number doesn't make a difference - when we take the derivative, the derivative is always 0.

2. Find $\frac{d}{dx} \int_3^x e^t \sin t dt$.

By FTC II, $\frac{d}{dx} \int_3^x e^t \sin t dt = e^x \sin x$

3. Find $\frac{d}{d\theta} \int_{-2}^{\theta} \arctan(x^2 + x) dx$

By FTC II, $\frac{d}{d\theta} \int_{-2}^{\theta} \arctan(x^2 + x) dx = \arctan(\theta^2 + \theta)$

4. Find $\int_1^{x^2} \cos t dt$ using the FTC I. Then take the derivative. What do you get?

By FTC I, $\int_1^{x^2} \cos t dt = \sin t \Big|_1^{x^2} = \sin(x^2) - \sin(1)$ Now we take the derivative: by the chain rule, $\frac{d}{dx} \sin(x^2) - \sin(1) = \cos(x^2)(2x) - 0$.

This example has a twist: our limit of integration is x^2 , not a plain x . This means when taking the derivative, we need to use the chain rule. (Also, when taking the derivative, $\sin(1)$ - or whatever the lower limit of integration is - is a number, so its derivative is 0.)

5. Find $\frac{d}{dx} \int_3^{\sin x} e^{-t^2} dt$

As in the last problem, we apply the chain rule as we use FTC II:

$$\frac{d}{dx} \int_3^{\sin x} e^{-t^2} dt = e^{-\sin^2 x} \cos x$$